

The Distribution of Wide Scope Indefinite Bare Duals in Kazym Khanty: a QUD-based Approach

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1. Introduction

It is often reported that, in languages without articles, bare NPs are unable to get wide scope existential (WSE) interpretations (Dayal 2004; Deal & Nee 2018; Srinivas 2021, a.m.o.). Generally this is given a semantic explanation. Bare NPs cannot get WSE readings either because they are definites (Dayal 2004 for non-plural bare NPs) or because they are frozen scope existentials (Chung & Ladusaw 2003; Srinivas 2021 for all bare NPs; Dayal 2004 for plurals). In this paper, I will argue that in at least some cases, the observed inability of bare NPs to get WSE readings is instead a side effect of pragmatics. I will base my argumentation on a case study from Kazym Khanty bare duals, which, in contrast to “classic” bare NPs, are variable-scope existentials (Tiutiunnikova et al. in press). And yet despite this, WSE uses are not freely available. I will argue, however, that this limitation has nothing to do with their scopal properties. The reason such WSE uses are ruled out is pragmatic. Namely, in such cases, bare duals are infelicitous because they cannot address a QUD that targets their not-at-issue content, namely the duality meaning. The case of Kazym Khanty bare duals shows that the absence of a WSE reading can only be considered as evidence of their scopal properties if the QUD parameter is controlled for.

This paper is organized as follows. In Section 2, I will discuss why Kazym Khanty data provide a challenge for the classic semantic accounts of bare NPs. In Section 3, I will review the scopal properties of Kazym Khanty bare duals. Then in Section 4, I will propose a pragmatic explanation for the limited distribution of the WSE uses. In Section 5, I will discuss the theoretical implications of the case study of Kazym Khanty, and will conclude in Section 6.

2. Bare NPs in articleless languages: issues with the WSE readings

Again, it is reported for many languages without articles that argumental bare NPs cannot get WSE readings. Consider an example from Teotitlán del Valle Zapotec (Deal & Nee 2018). In (1), the bare singular *beez* ‘frog’ is put into a context that forces the wide scope existential reading (‘there is a frog such that it is not in the basket’). However, using the bare singular in such a context leads to infelicity. To get the desired interpretation, the word *te* ‘one’ must be added.

- (1) [There’s six frogs and five of them are in the basket. We want to say there’s one that’s not, so we say:]

Kedih	y-u’-u-di	*(te)	beez	le’n	kanast
NEG	NEUT-be-NEG	one	frog	in	basket
‘One frog is not in the basket.’					(Deal & Nee 2018: 330)

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Similar cases are observed in other articleless languages, see, for instance, the work on Hindi (Dayal 2004 et cq.) and Kannada (Srinivas 2021). The usual explanations for this observation provided in the literature are *semantic*: bare NPs cannot get wide scope indefinite readings because they are not existential quantifiers in the first place. According to the most influential theory of nominal meaning in articleless languages, the neo-Carlsonian approach (Dayal 2004 et cq.), bare NPs in argumental positions receive either a definite (available for all bare NPs) or a narrowest scope existential interpretation (available only for bare plurals). There are some alternative analyses that derive the narrow scope existential interpretation for bare singulars as well (Chung & Ladusaw 2003; Srinivas 2021, a.m.o.), but crucially all of them predict that bare NPs cannot take variable scope, wide scope in particular¹.

Semantic analyses of this sort are unapplicable to Kazym Khanty data. As was shown in Tiutiunnikova et al. in press, bare singulars and bare duals in this language are variable-scope existentials. In particular, they can get WSE readings, as illustrated in (2). In the provided scenario, there are ten girls, and only two of them are not sleeping. Both the *narrow-scope existential* ('there are no two girls who are sleeping') and the *definite* ('the (only) two girls are not sleeping') interpretations of the bare dual *ewɛŋən* 'girls (two)' would contradict this scenario. Only the *wide scope existential* reading is compatible with it: it is indeed true that there are two girls who are not sleeping.

- (2) Context. In a camp, two out of ten girls cannot sleep. This was reported to a counselor Vasya, who then needed to calm them down. However, Vasya did not know the gender of the two children. Assuming those are boys, he takes soldier toys with him to play with the kids. On his way to the dorm, he meets a nurse who reported the issue, and she tells him:

Waśa,	ewɛŋən	ǎnt	uɫ-λ-əŋən,	akań	wuj-a.
V.	girl-DU	NEG	sleep-NPST-3DU	doll	bring

'Vasya, GIRLS are not sleeping, bring the dolls (instead).'

∃ > ¬

Interestingly, however, even though bare duals can take wide scope, such uses are limited in distribution. Compare (2) with (3), where the bare dual, being put in the same clause as in (2), is not able to get the WSE interpretation forced by the scenario.

- (3) Context. Masha works in the child summer camp, and her job is to check whether people are sleeping after midnight. There are ten girls and ten boys at the camp, and tonight, two girls were not sleeping. When the boss asked her "Are all the children sleeping?", she answered:

#antə,	ewɛŋən	ǎnt	uɫ-λ-əŋən
no	child-DU	NEG	sleep-PST-3DU

Intended: 'No, two girls are not sleeping.'

The unavailability of the WSE reading in cases like in (3) makes Kazym Khanty bare NPs look similar to their counterparts in other articleless languages. The problem is, however, that for Kazym Khanty bare duals, WSE readings *are* available in other cases. In this paper, I will argue that the oddity of (3) has nothing to do with the scopal properties of the bare NPs. Instead, (3) is bad for an independent pragmatic reason: the bare dual cannot address a QUD (Question Under Discussion; Roberts 1996 et cq.) that targets its not-at-issue content.

3. Kazym Khanty

3.1. Basic facts

Kazym Khanty is a dialect of Northern Khanty (Uralic family) spoken in the village Kazym in Khanty-Mansi Autonomous Okrug, West Siberia, Russia.

¹ The only exception is, technically, the hypothesis proposed in (Heim 2011), according to which bare NPs in articleless languages are indefinites without non-uniqueness (anti-)presupposition. The issue with this analysis is that it was proposed just as a null hypothesis (in languages without articles, bare NPs should be available in all indefinite and definite contexts), without any discussion of the relevant data. In particular, there is no explanation for why the WSE reading of bare NPs is unavailable in cases like in (1).

Nouns in Kazym Khanty feature a tripartite number system: singular, dual, and plural. The dual number is used when the number of the NP's satisfiers equals two.

- (4) a. *ńawrem* [child] 'child'
 b. *ńawrem-ηən* [child-DU] '(exactly two) children'
 c. *ńawrem-ət* [child-PL] 'children'

Kazym Khanty is an articleless language, and it uses bare NPs in argumental positions. Non-plural bare NPs in Kazym Khanty capture both definite and indefinite contexts and share all the semantic features of (non-restricted) existential quantifiers (Tiutiunnikova et al. in press). In the next section, I will review the main diagnostics from the paper, applying them to bare duals in particular.

3.2. Existential uses of bare duals

As existential quantifiers, Kazym Khanty bare duals feature the following properties. First, they do not require uniqueness. This is shown in (5), where the two bare duals quantify over two different pairs of Americans.

- (5)

<i>met</i>	<i>siri</i>		amerikants-ηən	<i>juχət-s-əηən</i> ,
most	front		American-DU	come-PST-3DU
<i>met</i>	<i>jułta</i>	(<i>isi</i>)	amerikants-ηən	<i>juχət-s-əηən</i>
most	last	also	American-DU	come-PST-3DU

 'Two Americans came first and two (other) Americans came last.'

Second, they can take variable scope. We already saw a bare dual taking scope above the negation in (2). In (6), the bare dual takes narrow scope with respect to the adverbial *kašən χatλ* 'everyday'. As ensured by the right context, (6) is compatible with the scenario where each day, a new couple of dogs is barking.

- (6)

<i>tātā</i>	<i>kašən</i>	<i>χatλ</i>	amp-ηən	<i>χuraś-λ-əηən</i> .
here	every	day	dog-DU	bark-NPST-3DU

 'A couple of dogs barks here every day.
 {Yesterday, **one couple** barked, today, **another couple** is barking.}'

Thus, the ability of bare duals to "refer" non-uniquely and take variable scope suggests that they should be analyzed as existential quantifiers, similar to *a*-indefinites in English. However, such an analysis can not explain why the bare dual can not take a wide scope with respect to negation in (3). I argue that the infelicity of cases like in (3) is not semantic, but pragmatic in nature. Namely, in such cases, bare duals are put into the context where they are forced to address a QUD targeting the cardinality of their satisfiers, and they can not do that because their duality meaning is not-at-issue content. In the next section, I will provide independent evidence for such an analysis.

3.3. (Not-)at-issue content of bare duals

In this section, developing the ideas proposed in (Golosov 2024), I will argue that the the duality meaning of the dual number is its not-at-issue content.

Let us start with the question test: according to the definition of at-issueness, only at-issue content can address the QUD (Simons et al. 2010; Tonhauser 2012). Crucially, only the predicative part of the bare dual's meaning can help answering a question, but not the duality implication. For instance, the sentence in (7) can address the question about identity of the players, but not the question about cardinality of the children.

- (7) {Who is playing on the street?/#How many children are playing on the street?}

<i>kam-ən</i>	<i>ńawrem-ηən</i>	<i>jun-λ-əηən</i>
street-LOC	child-DU	play-NPST-3DU

‘Children (two) are playing on the street.’

The second distinction between at-issue and not-at-issue contents is in their ability to determine the relevant set of focus alternatives (Tonhauser 2012). Namely, when a syntactic phrase is focused, only its at-issue content is involved in the semantic contrast, while its not-at-issue content is either shared among the contrasting alternatives or completely ignored by focus, depending on its projection properties.

Applying this test to Kazym Khanty bare duals, we can once again see that only the nominal predicate, but not the duality meaning, is a part of their at-issue content. This is illustrated in (8), where the same sentence with a bare dual is used in two different scenarios. The two contexts trigger different sets of focus alternatives. In Context 1, the goal of the speaker is to assert that only the female part of the cohort was present at the class, so only the predicate of the bare dual is contrasted, while the number of girls is irrelevant. In Context 2, the speaker highlights that among sixteen students, only two girls come. In this case, both the gender and the number of the students who came are contrasted with alternatives.

Crucially, the bare dual can be used only in the first context, not in the second: the duality meaning cannot be contrasted.

(8) ^{ok} Context 1. Students, *two* girls and eight boys, have their first day at school, so teachers keep track of their attendance. All of them were in the first class, but in the second class, only *the girls* came.

The person teaching the second class calls the other teacher and says:

Context 2. Students, *eight* girls and eight boys, have their first day at school, so teachers keep track of their attendance. All of them were in the first class, but in the second class, only *two girls* came.

The person teaching the second class calls the other teacher and says:

ma χuś-am-a εwe-nəŋ tɔp oməs-λ
I at-POSS.1SG-DAT girl-DU only sit-NPST[3SG]

‘Only two girls are sitting in my class.’

Thus, the semantic content of bare duals is heterogeneous with respect to its ability to address a QUD: only the nominal predicate lexicalized by a bare dual, but not the duality meaning, is its not-at-issue content. In particular, as we saw already, bare duals can not occur in contexts where a QUD would target the cardinality of their satisfiers.

In the next section, I will argue that this is exactly the reason for the infelicity of (3). Namely, (3) is odd not because the bare dual can not take the wide scope, but because the context enforces a cardinality-oriented QUD the bare dual can not address.

4. Proposal

In this section, I will argue that the analysis of argumental bare duals as existential quantifiers, as in (9)², proposed in (Goloso 2024; Tiutiunnikova et al. in press) is correct, and that the infelicity of cases like in (3) is a side effect of an independent pragmatic constraint: bare duals can not address a cardinality-oriented QUD.

(9) $\llbracket \text{N-DU}_{\text{ARG}} \rrbracket = \lambda P. \exists x[f(x) \ \& \ \#_f(x) = 2 \ \& \ P(x)]$, where f is the predicate denoted by N

Let us look again at (3), repeated here as (10). As follows from the context, the main QUD this utterance addresses is the polar question ‘Are all the children sleeping?’, and the clause after the negative answer is elaboration that addresses its own QUD. Now, a natural follow-up is to mention how many children are not sleeping. However, the duality implication of the dual number is not-at-issue content and cannot address such a QUD. Instead, the only at-issue content of (10) is ‘the person who is not sleeping is a girl’. Hence, the only QUD compatible with this follow-up is ‘Whoever is not sleeping, do they belong to the set of girls or boys?’, which is a rhetorically odd question in the provided context.

² $\#_f(x) = 2$ is true if and only if the number of x atomic with respect to f equals two (Goloso 2024: 324).

(10) QUD: Are all the children sleeping?

a. *antə*

no

Natural follow-up QUD: ‘How many children are not sleeping?’

b. #**ewɛ-ŋən** ǎnt ul-λ-əŋən
child-DU NEG sleep-PST-3DU
‘girls (*two*) are not sleeping.’

In contrast, the sentence in (2), repeated here as (11), is felicitous because in this case, the QUD targeting the at-issue content of the bare dual (its nominal predicate) is rhetorically natural. Namely, the goal of the speaker is to inform the addressee that the two children that are not sleeping are girls, not boys. The cardinality of the children is not questioned, so the not-at-issue content of the bare dual is not targeted by the QUD.

(11) QUD: ‘Children of what gender are not sleeping?’

ewɛ-ŋən ǎnt ul-λ-əŋən
girl-DU NEG sleep-NPST-3DU
‘*Girls* are not sleeping.’

5. Discussion

As we discussed in the previous section, the limited distribution of bare duals in Kazym Khanty, including some cases where their wide scope indefinite reading is forced, can be independently explained by a pragmatic constraint, while they still can be analyzed as existential quantifiers. In this section, I will discuss the important theoretical consequences of this case study.

The main lesson is this: the reason a wide scope indefinite reading is unavailable in a particular sentence is not always *semantic*. It may be pragmatic. So if the pragmatic context is not controlled for, this diagnostic can provide a false negative. The sentence could be understood as a response to a QUD targeting the cardinality of the tested NP’s satisfiers; in this case, if the meaning of the NP’s number feature is not-at-issue (as it is the case for Kazym Khanty dual), the sentence will be infelicitous regardless of the scopal properties of the NP.

This new caveat is important for diagnosing the scopal properties of bare NPs cross-linguistically. Yet to the best of my knowledge, it was not considered in previous research. Analysts did not consider the possibility that the WSE uses of bare NPs might be unavailable due to the observed pragmatic constraint. This means that at least in some other studied articleless languages, bare NPs that were diagnosed as unable to get the WSE readings might in fact be able to do so once asked in a proper context.

The chance of the misdiagnosis is even higher for some case studies because of how the context was set for the elicited examples. Consider, for instance, the sentence (1) from Teotitlán del Valle Zapotec, repeated here in (12). Since the goal of the speaker is to say that *one* frog is not in the basket, the accommodated QUD likely targets the quantity of the frogs out of the basket. If that is the case, and if the meaning of the singular number in Teotitlán del Valle Zapotec is not-at-issue, the infelicity of (12) can be independently explained pragmatically and hence does not tell us anything about the scopal properties of the bare singular in question.

(12)[There’s six frogs and five of them are in the basket. We want to say there’s one that’s not, so we say:]

Plausible hypothetical QUD: ‘How many frogs are not in the basket?’

Kedih y-u’-u-di *(te) beez le’n kanast
NEG NEUT-be-NEG one frog in basket
‘One frog is not in the basket.’

To rule out the possibility of false negative result, the sentence in (12) should be put in the context where QUD targets only the predicative part of the bare NP’s meaning, like in (2). For instance, it could be tested as an answer to the question ‘What type of animal is not in the basket?’.

6. Conclusions

In this paper, I addressed the issue of scopal behavior of bare NPs cross-linguistically. I argued that at least in some languages, what can be seen as an absence of the WSE reading is in fact a side effect of an independent pragmatic constraint. I based my argumentation on data from Kazym Khanty bare duals, which can get WSE readings, but not freely. I argued that the limited distribution of the WSE uses has nothing to do with the scopal properties of bare duals. Instead, in such cases, bare duals are infelicitous because they are forced to address the QUD with their not-at-issue content.

The case study of Kazym Khanty bare duals provides an important lesson for studying the scopal behavior of bare NPs cross-linguistically. Namely, to diagnose whether a bare NP can get a WSE reading, the researcher needs to elicit the stimulus in a context where the QUD does not target not-at-issue content of the tested NP, to avoid a false negative result. This applies to the existing studies as well: it can very well be that at least some tested bare NPs were falsely diagnosed as unable to get WSE readings, and they can, in fact, get them once put in a proper context.

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